SIMULOS®

Simulation Governance Architecture — Simulation Governance Layer

SIMULOS® serves as the official OOF® simulation governance architecture used for simulation governance design, simulation space identification, simulation methodology structuring, reality representation governance, assumption governance, simulation execution governance, simulation output governance, simulation-to-reality correlation, simulation evidence governance, reliance boundary definition, lifecycle monitoring, and resimulation governance across AI systems, autonomous systems, robotics, digital twins, engineering systems, scientific models, critical infrastructure, space systems, organizations, and future simulation environments.

Simulation Governance Architecture (SIMULOS®) does not define simulation itself. SIMULOS® defines the governance architecture required to ensure that simulations remain explicitly identified, purpose-bounded, reality-aware, assumption-governed, structurally controlled, input-aware, scenario-governed, execution-traceable, output-bounded, reality-correlated, evidentially constrained, and lifecycle-governed throughout the complete simulation lifecycle.

SIMULOS® consists of 12 Core Parent Standards and 60 Core Modules forming the primary simulation governance backbone within the OOF® ecosystem.

Additional compatible standards, bridge standards, and specialized simulation governance standards may extend the architecture when new simulation governance spaces are identified. However, the standards listed below form the official core simulation governance structure of SIMULOS®. Canonical Definition

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Simulation Governance Architecture (SIMULOS®) is the simulation governance architecture that defines the structural and methodological conditions under which a simulation is explicitly identified, purpose-bounded, connected to represented reality, assumption-governed, modelled within controlled environments, supplied with governed inputs, tested through relevant scenarios, executed under defined conditions, interpreted through bounded outputs, correlated with reality, converted into legitimately bounded evidence, and continuously reconsidered as reality and simulation conditions change throughout the complete simulation lifecycle. Core Question

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How can simulations remain methodologically governed from the identification of what is being simulated through representation, assumptions, modeling, execution, outputs, reality correlation, evidential reliance, and eventual resimulation, replacement, or retirement? Governance Flow

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Simulation Object → Simulation Purpose & Scope → Reality Representation → Simulation Assumptions → Simulation Model & Environment → Simulation Inputs & Data → Simulation Scenario Design → Simulation Execution → Simulation Output Governance → Simulation-to-Reality Correlation → Simulation Evidence & Reliance Boundary → Simulation Lifecycle & Resimulation Core Parent Standards
1. Simulation Object Standard (SOS)

Governed Space: Simulation Object

Core Question: What exactly is being simulated, and how must the Simulation Object be identified, bounded, versioned, and distinguished? 2. Simulation Purpose & Scope Standard (SPSS)

Governed Space: Simulation Purpose & Scope

Core Question: Why is the simulation being performed, what must it cover, and what remains outside its simulation boundary? 3. Reality Representation Standard (RRS)

Governed Space: Simulation Reality Representation

Core Question: Which aspects of reality must the simulation represent, at what abstraction and fidelity, and which aspects may legitimately remain simplified or excluded? 4. Simulation Assumption Standard (SAS)

Governed Space: Simulation Assumptions

Core Question: Which assumptions enable the simulation, and how must their basis, dependencies, uncertainty, limitations, and consequences remain governed? 5. Simulation Model & Environment Standard (SMES)

Governed Space: Simulation Model & Environment

Core Question: How must the simulation model and simulation environment be constructed, configured, bounded, and controlled? 6. Simulation Input & Data Standard (SIDS)

Governed Space: Simulation Inputs & Data

Core Question: Which inputs and data may enter the simulation, and under what representativeness, transformation, temporal, and uncertainty conditions? 7. Simulation Scenario Standard (SSS)

Governed Space: Simulation Scenario Design

Core Question: Which scenarios must be simulated to represent expected conditions, variations, edge cases, stresses, uncertainty, and materially relevant alternatives? 8. Simulation Execution Standard (SES)

Governed Space: Simulation Execution

Core Question: How must a simulation be instantiated, executed, observed, repeated, controlled, and documented? 9. Simulation Output Governance Standard (SOGS)

Governed Space: Simulation Output Governance

Core Question: What exactly did the simulation produce, how must simulation outputs be bounded and interpreted, and what may legitimately be inferred from them? 10. Simulation-to-Reality Correlation Standard (SRCS)

Governed Space: Simulation-to-Reality Correlation

Core Question: To what extent do simulated behavior, conditions, and outputs correspond to the reality they claim to represent? 11. Simulation Evidence & Reliance Boundary Standard (SERBS)

Governed Space: Simulation Evidence & Reliance Boundary

Core Question: What evidential meaning may simulation results legitimately carry, and where does simulation-specific reliance end before broader validation governance becomes necessary? 12. Simulation Lifecycle & Resimulation Standard (SLRS)

Governed Space: Simulation Lifecycle, Drift & Resimulation

Core Question: How must simulation relevance be monitored over time, and when must changed reality trigger update, resimulation, replacement, or retirement? Compatible OOF® Architectures

SIMULOS® interoperates with multiple OOF® architectures governing complementary governance spaces.

GOA™ — Governance Architecture provides the constitutional governance layer defining how governance itself is structured, bounded, delegated, and maintained.

ORA™ — Operational Reality Architecture governs what is actually occurring in Operational Reality, while SIMULOS® governs how selected aspects of that reality are represented and simulated.

VALIDOS® — Validation Governance Architecture governs whether the total validation basis sufficiently supports a Validation Determination and intended reliance, while SIMULOS® determines what simulation and simulation-derived evidence can legitimately represent and support.

INTEGROS® — Integrity Governance Architecture governs integrity across objects, states, evidence, and processes, while SIMULOS® governs simulation-specific structure, methodology, execution, representation, correlation, evidence boundaries, and lifecycle.

ADIT® governs evidential observation, preservation, verification, and auditability of what occurred, while SIMULOS® governs simulation execution itself and the simulation-specific meaning of its resulting outputs and evidence.

AGA™ — Accountability Governance Architecture governs authority, responsibility, accountability, attribution, and decision relationships surrounding simulation activities and reliance.

AIG® — AI Governance Architecture governs the behavioral governance lifecycle of actual AI systems that may be represented, tested, or evaluated through simulation.

ASGA™ — Autonomous Systems Governance Architecture governs actual autonomous systems, agents, authority, constraints, and autonomous operational behavior that may be represented or evaluated through simulation.

SIMULOS® serves as the common simulation governance architecture supporting the broader OOF® ecosystem by ensuring that simulations remain methodologically connected to the objects, realities, assumptions, conditions, executions, outputs, evidence, and lifecycle states they claim to represent.
Architectural Position

SIMULOS® governs the complete simulation governance lifecycle.

The architecture progresses through:

Simulation Object → Simulation Purpose & Scope → Reality Representation → Simulation Assumptions → Simulation Model & Environment → Simulation Inputs & Data → Simulation Scenario Design → Simulation Execution → Simulation Output Governance → Simulation-to-Reality Correlation → Simulation Evidence & Reliance Boundary → Simulation Lifecycle & Resimulation

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Together these standards govern how a simulation target is identified, why simulation is undertaken, how reality is represented, which assumptions make the simulation possible, how models and environments are constituted, which inputs enter the simulation, which scenarios are tested, how execution occurs, what outputs are produced, how those outputs correspond to reality, what evidential meaning may legitimately be derived from them, and when changed reality requires resimulation, replacement, transition, or retirement.

Unlike a linear simulation methodology, SIMULOS® becomes recursive after operational use begins.

The extended lifecycle becomes:

Simulation → Operational Use → Reality Change → Simulation Drift → Resimulation Trigger → Resimulation Scope → Lifecycle Re-Entry → Resimulation / Replacement / Retirement

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This enables a governed simulation to remain connected not merely to the reality represented when it was created, but also to the reality that continues to change after simulation results begin influencing operational decisions. Core Governed Simulation Spaces

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SIMULOS® establishes twelve distinct governable simulation spaces:

Simulation Object → what exactly is being simulated.

Simulation Purpose & Scope → why the simulation exists and what it must and must not cover.

Simulation Reality Representation → which parts of reality are represented, simplified, abstracted, or excluded.

Simulation Assumptions → which assumptions enable the simulation and how their consequences remain visible.

Simulation Model & Environment → how the model and simulation environment are structured, configured, and controlled.

Simulation Inputs & Data → what enters the simulation and under which representativeness, transformation, temporal, and uncertainty conditions.

Simulation Scenario Design → what conditions, variations, edge cases, stresses, and alternatives are actually tested.

Simulation Execution → how simulations are instantiated, controlled, observed, repeated, and documented.

Simulation Output Governance → what the simulation produced and what may legitimately be inferred from those outputs.

Simulation-to-Reality Correlation → how simulated behavior, conditions, and outputs correspond to observed or governed reality.

Simulation Evidence & Reliance Boundary → what simulation-derived evidence may legitimately mean and where simulation-specific reliance must stop.

Simulation Lifecycle, Drift & Resimulation → how continuing relevance is monitored and when change requires resimulation, replacement, transition, or retirement.

Each space answers a different methodological question.

Each can fail independently.

Their separation is therefore foundational to the SIMULOS® architecture.
Cross-Architecture Boundary Lock

SIMULOS® is intentionally bounded to prevent duplication with adjacent OOF® architectures. SIMULOS® ↔ ORA™

ORA™: What is actually happening in Operational Reality?

SIMULOS®: How is selected reality represented and simulated?

Operational Reality ≠ Simulated Representation of Reality. SIMULOS® ↔ VALIDOS®

SIMULOS®: What does the simulation and simulation-derived evidence legitimately represent and support?

VALIDOS®: Does the total validation basis sufficiently support the Validation Determination and intended reliance?

Simulation Evidence ≠ Validation Determination. SIMULOS® ↔ INTEGROS®

SIMULOS®: governs simulation-specific structure, representation, assumptions, execution, outputs, correlation, evidence boundaries, and lifecycle.

INTEGROS®: governs integrity across objects, states, evidence, and processes.

Simulation Governance ≠ General Integrity Governance. SIMULOS® ↔ ADIT®

SIMULOS®: governs simulation execution and the simulation-specific structures through which simulation activity occurs.

ADIT®: governs evidential observation, preservation, verification, examination, and auditability of what occurred.

Simulation Execution ≠ Audit of Simulation Execution. SIMULOS® ↔ AGA™

SIMULOS®: governs simulation methodology and simulation-specific conditions.

AGA™: governs who holds authority, responsibility, accountability, and attribution for simulation-related decisions and failures.

Simulation Methodology ≠ Accountability Allocation. SIMULOS® ↔ AIG® / ASGA™

SIMULOS® may simulate AI systems, autonomous agents, robots, and autonomous operational environments.

SIMULOS®: governs their simulated representation and simulation lifecycle.

AIG® / ASGA™: govern the behavior, autonomy, authority, constraints, and operation of the actual systems.

Simulating an autonomous system ≠ governing the autonomy of the actual system.
Architectural Origin

SIMULOS® evolved from the foundational:

SIMULOS® — Simulation Integrity Standard

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which established the original methodological concern that simulation must remain structurally connected to the reality, assumptions, conditions, execution basis, and outputs it claims to represent.

As the governed space expanded, simulation integrity was identified as only one part of a substantially larger governance problem.

The architecture therefore evolved from a single integrity-oriented standard into a complete simulation governance architecture.

Its foundational principles are distributed primarily across:

Reality Representation → Simulation Assumptions → Simulation Model & Environment → Simulation Execution → Simulation Output Governance → Simulation-to-Reality Correlation → Simulation Evidence & Reliance Boundary

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The original standard remains part of the architectural history of SIMULOS® as its Foundational Reference Standard, while the twelve Parent Standards constitute the current canonical architecture. Architectural Structure

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- Architecture: SIMULOS® — Simulation Governance Architecture
- Category: Governance & Enforcement
- Subcategory: Simulation Governance
- Core Parent Standards: 12
- Core Modules: 60
- Architecture Model: Lifecycle-Based · Modular · Recursive
- Primary Governance Domain: Simulation Governance

The architecture deliberately separates governable spaces that can independently fail.

A simulation may:

- represent the wrong object; - pursue an unclear purpose; - represent reality inadequately; - depend upon unsupported assumptions; - use an inappropriate model; - consume unrepresentative inputs; - omit decisive scenarios; - execute incorrectly; - produce misinterpreted outputs; - correlate poorly with reality; - generate evidence used beyond its legitimate meaning; - remain operational after reality has materially changed.

SIMULOS® treats these not as one generic simulation quality problem, but as distinct governance spaces requiring distinct methodological controls. Recursive Lifecycle Principle

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SIMULOS® does not end when a simulation run ends.

A simulation may remain operationally influential for months, years, or decades after execution.

During that period:

Reality may change. The Simulation Object may change. Assumptions may become obsolete. Inputs may shift. New scenarios may emerge. Models may become outdated. Correlation may deteriorate. Evidence may lose applicability.

SIMULOS® therefore closes recursively through SLRS:

Operational Use → Reality Change → Simulation Drift → Resimulation Trigger → Resimulation Scope → Lifecycle Re-Entry → Resimulation / Replacement / Retirement

This makes lifecycle change part of simulation governance rather than an external maintenance concern. Foundational Architecture Principles

A simulation cannot legitimately represent an object more precisely than that object has been identified and bounded.

Simulation purpose determines what the simulation must represent, not what the technology happens to be capable of simulating.

Represented reality is not Operational Reality.

Simulation similarity is not reality equivalence.

An assumption that remains hidden cannot remain governed.

Model sophistication does not compensate for an incorrectly represented problem.

More data does not correct an incorrectly bounded simulation.

Scenario volume is not scenario coverage.

Execution precision does not guarantee interpretive legitimacy.

Simulation output is not automatically simulation evidence.

Evidence volume is not evidential strength.

Simulation-derived evidence must not travel farther than its governed meaning can legitimately support.

A simulation that was once relevant does not remain relevant merely because it continues to execute.

Changed reality may require changed simulation.

Resimulation is part of simulation governance, not an admission that the original simulation failed. Canonical Principle

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SIMULOS® does not govern Operational Reality itself, general validation sufficiency, general integrity, auditability, accountability, or the behavior and authority of actual AI and autonomous systems. SIMULOS® governs the methodological conditions under which reality is represented through simulation, simulation activity is structured and executed, simulation outputs acquire bounded meaning, simulation-derived evidence acquires legitimate reliance boundaries, and continuing changes in reality trigger governed resimulation, replacement, transition, or retirement throughout the complete simulation lifecycle.

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